THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 7: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A die is rolled. The probability of getting a prime number or a multiple of 3 is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{2}$

2. A bag has 4 red and 6 blue balls. After adding 6 more red balls, the probability of drawing a red ball is: 1

(a) $\frac{3}{8}$ **(b)** $\frac{5}{8}$ **(c)** $\frac{5}{9}$ **(d)** $\frac{3}{4}$

3. A letter is chosen at random from the word MATHEMATICS. The probability that it is the letter M or T is: 1

(a) $\frac{5}{11}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{7}{11}$ **(d)** $\frac{4}{11}$

4. In a town of 8000 people the probability that a randomly chosen person owns a smartphone is $\frac{7}{10}$. The expected number of owners is: 1

(a) 5602 **(b)** 5601 **(c)** 5600 **(d)** 5599

5. If $P(E) = 7/15$, then the probability that E does not happen is: 1

(a) $\frac{8}{15}$ **(b)** $\frac{7}{15}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{3}{5}$

6. Three coins are tossed. The probability of getting the same face on all three coins is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{1}{5}$

7. Two dice are rolled together. The probability that the sum is 4 is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{1}{13}$ **(c)** $\frac{11}{12}$ **(d)** $\frac{1}{12}$

8. Two dice are rolled together. The probability of getting a sum of 12 is: 1

(a) $\frac{35}{36}$ **(b)** $\frac{1}{18}$ **(c)** $\frac{1}{36}$ **(d)** $\frac{1}{37}$

9. A die is rolled. The probability of getting an even number or a multiple of 3 is: 1

(a) $\frac{3}{4}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{2}{3}$

10. A bag has 5 red and 7 blue balls. After adding 3 more red balls, the probability of drawing a red ball is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{7}{15}$ **(c)** $\frac{3}{5}$ **(d)** $\frac{8}{15}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A bag has 5 red and 10 blue balls. 5 more red balls are added. Find the new probability of drawing a red ball. 2

12. The probability of winning a game is $\frac{3}{8}$. Find the probability of not winning, and the expected number of wins in 160 games. 2

13. Two dice are rolled together. Find the probability that the sum of the numbers is 10. 2

14. A bag has 7 red and 5 blue balls. 2 more red balls are added. Find the new probability of drawing a red ball. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A die is rolled. Find the probability of getting 3
(i) an even number,
(ii) a multiple of 3,
(iii) an even number or a multiple of 3.

16. In a survey of 120 people, 40 like tea only, 30 like coffee only and 20 like both. Find the probability that a person chosen at random likes tea (tea only or both), and the probability that the person likes neither (the rest). 3

17. A bag has 5 red and 15 blue balls. How many red balls must be added so that the probability of drawing a red ball becomes $\frac{3}{4}$? 3

OR

A die is rolled. Find the probability of getting 3
(i) a perfect square,
(ii) a number at least 5,
(iii) a perfect square or a number at least 5.

18. In a survey of 250 people, 90 like tea only, 70 like coffee only and 35 like both. Find the probability that a person chosen at random likes tea (tea only or both), and the probability that the person likes neither (the rest). 3

OR

A bag has 2 red and 10 blue balls. How many red balls must be added so that the probability of drawing a red ball becomes $\frac{1}{2}$? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Two dice are rolled together. Find the probability of 5
(i) a sum of 6,
(ii) a sum of 8,
(iii) a sum of 6 or 8.

OR

A factory tested 250 bulbs and found 10 defective. Find the probability that a bulb is defective, and estimate the number of defective bulbs in a lot of 7500. 5

20. Two dice are rolled together. Find the probability of 5
(i) a sum of 2,
(ii) a sum of 12,
(iii) a sum of 2 or 12.

OR

A factory tested 400 bulbs and found 12 defective. Find the probability that a bulb is defective, and estimate the number of defective bulbs in a lot of 16000. 5

END OF QUESTION PAPER Total: 40 Marks

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